

ECE131a Extra Credit Project

Distributed Detection in Gaussian Noise

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1 Introduction

1.1 Background

This problem is adapted from the results in [1] and [2]. A scalar signal $S \in \{-m, m\}$ is transmitted from a satellite to a field of n sensors on the surface of the earth. S communicates an important message, and is equally likely to take either value so that

$$P_S(s) = \begin{cases} \frac{1}{2}, & \text{for } s = -m, \\ \frac{1}{2}, & \text{for } s = m, \\ 0, & \text{otherwise.} \end{cases}$$

Sensor i observes random variable

$$X_i = S + Z_i$$

where the random variables Z_i are independent and identically distributed according to a standard normal distribution $\mathcal{N}(0, 1)$. This project explores how to efficiently and reliably estimate S from the vector of observations

$$\mathbf{X} = [X_1, X_2, \dots, X_n].$$

Any estimate $\hat{S}$ will naturally depend on the observables. A *statistic* is a function of the set of observables that we use in preparing our estimate. For this problem we will be especially interested in the statistic

$$T_n = \frac{1}{n} \sum_{i=1}^n X_i,$$

which is sometimes called the sample mean.

A *sufficient statistic* of a set of observations for estimation of a particular random variable is one that allows as good of an estimate of the random variable as the original observations themselves. T_n is a sufficient statistic of $\mathbf{X}$ for S if and only if

$$f_{X|T,S}(x|t, s) = f_{X|T}(x|t).$$

In our specific case, this turns out to be true so that the sample mean is a sufficient statistic of $\mathbf{X}$ for S . We can use the sample mean T_n to estimate S and be confident that we have not lost any information. Specifically, we will estimate S as follows:

$$\hat{S} = \begin{cases} m, & \text{if } T_n \geq 0, \\ -m, & \text{otherwise.} \end{cases}$$

1.2 Report Organization

The project is 25 project points which is equivalent to 5% of the total grade percentage as extra credit. The project must be completed to earn an A+ grade.

There are 8 problems, each answer must have a Matlab plot to back up the answer.

Project Statement:

ECE 131a Spring 2026 Extra Credit Project, Distributed Detection in Gaussian Noise

2 Problem 1

3pts.

Suppose that $m = 0.5$. Use Matlab function `semilogy` to plot the probability of error $P(\hat{S} \neq S)$ as a function of the number of sensors n for $1 \leq n \leq 100$. The Matlab function `qfunc` is your friend for this problem. How many sensors do you need so that $P(\hat{S} \neq S) < 10^{-3}$? Call this number n_g (as in the number of Gaussian samples).
